

Hyperkalemia

The three-phase approach: stabilize, shift, remove

Calcium stabilizes the myocardium — it does NOT lower the potassium.

Why we get consulted

The urgency comes from the arrhythmia risk, not the number.

TRIGGERS FOR CONSULT

- K $\geq$ 6.0 or any ECG changes
- Refractory despite medical therapy
- ESRD / CKD-5 needing KRT
- Hyperkalemia in AKI
- Recurrent on RAAS / MRA
- Binder selection (patiromer / SZC)

TEACHING PEARL

Three questions in order: is it real? Is the myocardium at risk (ECG)? What's the mechanism? Then treat. Skipping the ECG is the classic error.

First question: is this real?

Rule out pseudohyperkalemia before you commit to aggressive therapy.

PSEUDOHYPERKALEMIA — consider if

- Hemolyzed sample (difficult draw, tourniquet)
- Fist clenching during phlebotomy
- Platelets > 400K or WBC > 50K (cells release K⁺ in tube)
- Delayed transport (cold → K⁺ leak)
- No ECG changes despite reported K⁺ 7+

Action: redraw (plasma, not serum if thrombocytosis); repeat ECG

TRUE HYPERKALEMIA — drivers

↓ Excretion: AKI, CKD 4–5, RAAS, MRA, K-sparing diuretics, heparin, calcineurin inhibitors, trimethoprim-sulfamethoxazole (TMP-SMX)

Cellular shift: acidosis, insulin deficiency (DKA), β-blockade, succinylcholine, digoxin toxicity

↑ Load: tumor lysis, rhabdo, massive transfusion, K⁺ supplements, salt substitutes

Hypoaldo: type 4 RTA (DM), adrenal insufficiency

Action: treat the mechanism, not just the number

ECG: the driver of urgency

ECG changes correlate poorly with K^+ level — but any change means membrane stabilization now.

K^+ range	Typical ECG	Action
5.5 – 6.0	Often none; occasional peaked T	Hold RAAS/MRA; diet; consider binder
6.0 – 6.5	Peaked T waves (narrow-based, tall)	Shift therapy; binder; reassess every 2 h
6.5 – 7.5	PR prolongation, QRS widening, loss of P	Calcium NOW; shift; strongly consider KRT
> 7.5	Sine wave, VT/VF, asystole	Calcium, shift, emergent KRT

Classic teaching error: waiting to treat until ECG changes appear. Some patients with K^+ 6.0 have widened QRS; others with K^+ 7 look normal. Treat based on rate of rise, AKI context, and ECG together — not K^+ value alone.

Three phases — don't mix them up

Phase 1 protects the heart. Phase 2 buys time. Phase 3 removes the potassium.

1

STABILIZE

Protect the myocardium. Buys 30–60 minutes. Does NOT lower K^+ .

2

SHIFT

Move K^+ intracellularly. Drops K^+ 0.5–1.0 mEq/L for 2–6 h. Does NOT remove K^+ .

3

REMOVE

Excrete K^+ from the body. The only definitive step.

Mixing phases is the classic trainee error: giving calcium and stopping, or giving insulin without a definitive removal plan.

Drugs, doses, and kinetics

Know onset and duration — it tells you when to re-check and re-dose.

Drug	Dose	Onset / Duration	Notes
Ca gluconate 10%	10 mL IV over 2–5 min (can repeat ×1)	1–3 min / 30–60 min	<i>STABILIZE only. Avoid peripheral Ca chloride (tissue necrosis). Caution on digoxin.</i>
Insulin + dextrose	10 U reg IV (or 5 U if eGFR < 30) + 25 g D50	10–20 min / 4–6 h	<i>Hypoglycemia in ~20–30% at 10 U in renal failure; glucose q1h × 6 h.</i>
Albuterol	10–20 mg nebulized (4–8× the asthma dose)	30 min / 2–4 h	<i>Additive with insulin. Avoid in active ischemia.</i>
NaHCO₃	150 mEq in D5W over hours (if acidosis)	Hours	<i>ONLY if metabolic acidosis — not a primary K⁺ lowerer.</i>
Furosemide	40–80 mg IV if UOP present	30–60 min / 4–6 h	<i>Removes K⁺ via urine; useless if anuric.</i>
Patiromer / SZC	Patiromer 8.4 g PO daily • SZC 10 g TID × 48 h	Patiromer ~7 h • SZC ~1 h	<i>GI binders. Outpatient control on RAAS. Not for life-threatening hyperK.</i>
Hemodialysis	K 2.0 or 1.0 bath × 3–4 h	Minutes / hours	<i>Definitive. Removes 25–50 mEq/session. Expect rebound in 2–4 h.</i>

Urgent hyperkalemia algorithm

From the moment you're paged: this is what happens in order.

- 1** **Get the ECG immediately** Peaked T, wide QRS, or any change → emergency tier.
- 2** **Stabilize if ECG changes OR $K \geq 6.5$** Calcium gluconate 1 g IV. Re-dose if ECG not improved in 5 min.
- 3** **Shift** Insulin 10 U + D50 25 g IV. Albuterol 10–20 mg neb. Bicarb if pH < 7.20.
- 4** **Remove** Loop diuretic if UOP present. Binder (SZC/patiromer). Call dialysis if refractory or anuric.
- 5** **Fix mechanism, monitor** Hold RAAS/MRA/ K^+ supplements. Repeat K^+ + ECG q1–2 h. Expect rebound in 2–4 h.

Outpatient: keep the RAAS on

Modern binders let us continue ACEi / ARB / finerenone / spironolactone in CKD.

BEFORE STOPPING RAAS

Re-check K^+ — fasting, non-hemolyzed sample

Low- K^+ diet counseling (not salt substitutes!)

Diuretic if volume-tolerant (loop $\pm$ thiazide)

Treat metabolic acidosis: HCO_3^- if $< 18-20$

Stop K-sparing supplements, NSAIDs, bactrim

Only after this — try a binder

BINDERS — PATIROMER vs SZC

Patiomer (Veltassa): 8.4–25.2 g PO daily. Takes 7 h for onset. Separate from other meds by 3 h. Constipation, $\downarrow Mg^{2+}$.

Sodium zirconium (Lokelma): 10 g TID $\times$ 48 h, then 10 g daily. Faster (1 h). High sodium load (caution HF).

Both permit continuation of RAAS / MRA in CKD (AMBER trial — patiomer; DIAMOND — patiomer in HFrEF).

Kayexalate (sodium polystyrene sulfonate (SPS)): avoid — risk of bowel necrosis; use SZC instead.

Common pitfalls

If you avoid these, you'll stay out of trouble.

AVOID

- Forgetting the ECG
- Assuming calcium lowered K
- Bicarb without acidosis
- Loop in anuria
- Kayexalate when SZC available
- Stopping RAAS permanently
- Missing insulin hypoglycemia

DO INSTEAD

- Always present the ECG
- State: 'calcium stabilized; K still X'
- Check bicarb before NaHCO_3
- Anuria → dialysis for removal
- SZC / patiromer, not Kayexalate
- Diet + binder before stopping RAAS
- Glucose q1h × 6 h after insulin

Clinical case

Think it through — answer is on the next slide.

VIGNETTE

A 68-year-old woman with T2DM, CKD 3b (eGFR 35), HFrEF 30%, on lisinopril 40 mg, spironolactone 25 mg, metoprolol, and furosemide 40 mg. ER check-in K⁺ 6.8 after missing a week of follow-up. BP 138/80, HR 62. ECG: peaked T waves, PR 220, QRS 105 ms. Pt is asymptomatic.

QUESTION: What's the immediate sequence of interventions, and what's your outpatient plan?

Clinical case — answer

Key teaching points from the case.

ANSWER

Calcium gluconate 1 g IV now, then shift (insulin 10 U + D50 25 g; albuterol 10 mg neb), then remove (loop diuretic if UOP, SZC 10 g TID).

TEACHING

Three phases in order: STABILIZE → SHIFT → REMOVE. Calcium protects the heart but does NOT lower K⁺. After shift, the K⁺ rebounds in 4-6 h. Outpatient: her RAAS + MRA + CKD + diabetes are all driving hyperK, but these drugs are prognostically essential. Rather than stopping them permanently, add SZC or patiomer (AMBER, DIAMOND) and continue RAAS/MRA at adjusted doses. Diet counseling (avoid salt substitutes, bananas, potatoes) and adherence check are equally important.

REFERENCES

Hyperkalemia

Guidelines

KDIGO 2024 CKD Guideline (potassium management)

ESC/AHA: hyperkalemia in HFrEF on MRA

ERA-EDTA position statement on hyperkalemia management

KDOQI commentary on dietary K⁺ in CKD

Landmark Trials

PEARL-HF — patiromer in HFrEF on MRA

OPAL-HK — patiromer for RAAS continuation

AMBER — patiromer in resistant HTN + CKD on spiro

DIAMOND — patiromer enables MRA in HFrEF

HARMONIZE — SZC maintenance of normokalemia

UpToDate (2026 access):

Causes and evaluation of hyperkalemia in adults • Clinical manifestations of hyperkalemia in adults • Treatment and prevention of hyperkalemia in adults